

CAREER OBJECTIVE

A passionate Electronics and Communication Engineering graduate seeking a fresher role in Embedded Systems, aiming to apply hands-on experience in Embedded C, ARM microcontrollers, and real-time systems to contribute effectively to an organization while continuously growing as an embedded engineer.

EDUCATION

Course	Institution	Board/University	Year	Percentage
B.Tech in ECE	SVR ENGINEERING COLLEGE, NANDYAL	JNTUA	2025	67.30
Diploma in ECE	ESC GOVT. POLYTECHNIC COLLEGE, NANDYAL	APSBTET	2022	67.97
SSC	ZPHS, RAJANAGARAM	STATE BOARD OF SECONDARY EDUCATION, AP	2019	87.40

TECHNICAL SKILLS AND INTERESTS

Programming Languages: C, Embedded C, C++, Data Structures

Embedded & OS: ARM, Linux, RTOS

Microcontrollers: LPC2148

Networking: TCP/IP

Tools & Concepts: Keil uVision, Microcontroller Programming, Debugging, Problem Solving

SOFT SKILLS

- Strong analytical and problem-solving skills
- Clear and effective communication
- Teamwork and time management
- Quick learner with strong attention to detail

PROJECTS

AI - POWERED SMART NOTICE BOARD USING RASPBERRY PI & CHATBOT INTEGRATION

DESCRIPTION:

- Designed and developed a smart digital notice board using Raspberry Pi capable of displaying dynamic content remotely.
- Integrated a RASA-based AI chatbot to enable interactive voice and text-based responses for users.
- Implemented Wi-Fi connectivity for real-time remote content updates without manual intervention.
- Built the entire application using Python, handling both the chatbot logic and display rendering.
- Enabled automated notice management, reducing dependency on manual notice board updates.

Tech Stack: Python, Raspberry Pi, RASA Chatbot, Wi-Fi.

Duration: 3 months

MENU-DRIVEN BANKING SYSTEM IN C USING SINGLY LINKED LIST AND FILE HANDLING

DESCRIPTION:

- Developed a modular banking application in C supporting account creation, deposit, withdrawal, and money transfer.
- Implemented dynamic account management using singly linked lists.
- Stored and restored account data using binary file handling (fwrite, fread).
- Maintained last five transaction records per account.

CANTRACK: EMBEDDED AUTOMOTIVE PARAMETER DISPLAY CONSOLE

DESCRIPTION:

- Developed a 3-node distributed embedded system using CAN protocol for ECU communication.
- Implemented message framing, CAN ID configuration, and Interrupt based resumption.
- Configured and handled External interrupts (EINT0, EINT1) for indicator toggle control.
- Integrated DS18B20 Temperature sensor using 1-wire protocol.
- Implemented LED running pattern for automotive-style left and right indicator simulation.

Tech Stack: Embedded C, LPC2148, CAN Protocol, DS18B20, 1-Wire Protocol.

POWER CHRONO – TIME-DRIVEN ENERGY AUTOMATION SYSTEM

DESCRIPTION:

- LPC2148, Embedded C, RTC, LCD, Keypad, External Interrupts. Designed and implemented a real-time energy automation system to control electrical loads based on RTC schedules.
- Developed a menu-driven user interface using LCD and keypad for editing RTC values and configuring ON/OFF timings.
- Built modular Embedded C drivers for LCD, RTC to ensure efficient real-time performance.
- Achieved reliable automatic load control, improving overall power management efficiency.

CERTIFICATIONS

EduSkills – Android Developer Virtual Internship | *April - June 2024*

NPTEL – Certified in Internet of Things | *November 2024*

Vector India, Hyderabad – Completed a 6 month course in Embedded Systems | *June 2025 - January 2026*

Cisco – Completed a 3 month online training on IT Essentials: PC Hardware and Software Course during my diploma final year. | *Feb - April 2022*

AREA OF INTEREST

- Embedded Systems
- Programming
- Hardware Design & Analysis
- Internet of Things

HOBBIES

- Travelling
- Knowing about new things
- Playing sports

DECLARATION

I hereby declare that the above written particulars are true to my knowledge and belief.

Place: Hyderabad

SUKAVASI VINAY KUMAR